

Predicting FictiPay Customer Churn

An end-to-end behavioral pipeline at 73M-transaction scale

0.985

OOF AUC-ROC

~70

engineered features

4

model families blended

12.7%

churn base rate

Churn = no transaction in April · features built only from Jan–Mar · evaluated on AUC-ROC

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Scale → Features → Ensemble

1 · Out-of-core data

- **Polars streaming** over 73M transactions + 77M balances
- Columnar projection, per-month partitions, feature caching
- Runs in 16 GB RAM — no cluster needed

2 · RFMC features

- Recency, Frequency, Monetary, **Clumpiness**
- Balance dynamics, P2P network, type-mix, KYC
- Multi-window (7/30/90d) + weekly/daily sequence

3 · Models

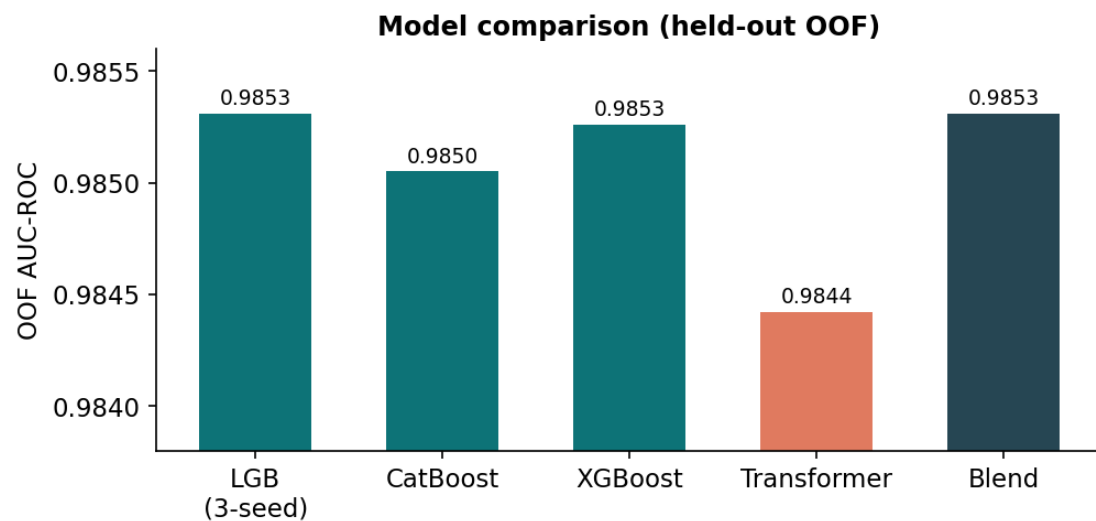
- LightGBM (Optuna-tuned, 3-seed)
- XGBoost + CatBoost (GPU)
- Per-event **Transformer** on raw sequences
- Rank-blended for low variance

Imbalance

- Class-weighted loss (no SMOTE — harmed the tail)

RESULTS

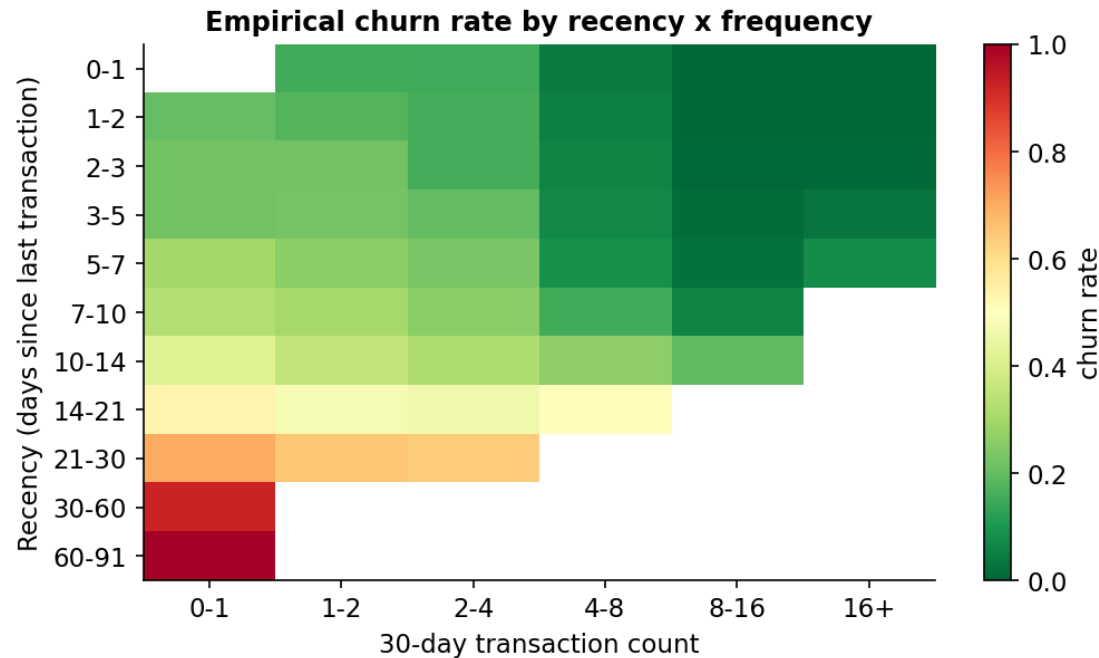
0.985 OOF — and a clear ceiling



Metric	Value
AUC-ROC	0.9853
Average Precision	0.916
Precision @ top 10%	0.910
Recall @ top 10%	0.717

Four independent model families converge at ~0.985. 81% of accounts are active to the window edge — their April churn is near-random, the behavioral ceiling. Train/test identically distributed (adversarial AUC 0.52).

What drives churn



Top drivers (SHAP)

- **Recency** — days since last activity dominates
- **30-day frequency** — strongest protective factor
- **Late-window share** — edge activity predicts continuation
- **counter-intuitive Frozen balance** — recent transaction but stalling balance still churns: silent disengagement

Leakage audit

ID order, TrxID, timestamps, April bleed, P2P neighbours — all probed, all clean. Every driver is genuine behavior.

Flag the top 10% — act before they leave

Decision rule

Target the highest-risk decile each cycle. At 0.91 precision in that decile, **~9 in 10 flagged users are true churn risks**, capturing ~72% of all churners — efficient retention spend.

Why it pays

Retaining a wallet user is far cheaper than reacquiring one; concentrating offers on a precise decile maximizes ROI.

Interventions

- **Frozen-balance users** → cash-in incentive / fee waiver
- **Lapsed bill-payers** → autopay nudge before due date
- **Declining frequency** → personalized cashback on their top transaction type
- **Shrinking network** → referral & P2P reward

Monitoring

Score weekly; watch balance-change frequency as an early-warning signal.